Steel Beam Check: W18X50

Project	Example	Member	B-1
Engineer	-	Date	2026-10-22
Method	LRFD (1.2D + 1.6L)	Software	Beam Check Desk v0.1.0

PASSES. Governing check: Flexure (1.2D + 1.6L), demand/capacity = 0.632.

Inputs

Beam length	24 ft	Supports	Simple supports at 0 ft and 24 ft
Fy	50 ksi	E	29000 ksi
Unbraced length Lb	0 ft (continuously braced)	Cb	1.00
Self weight	included (50 lb/ft)	Deflection limits	L/360 live, L/240 total

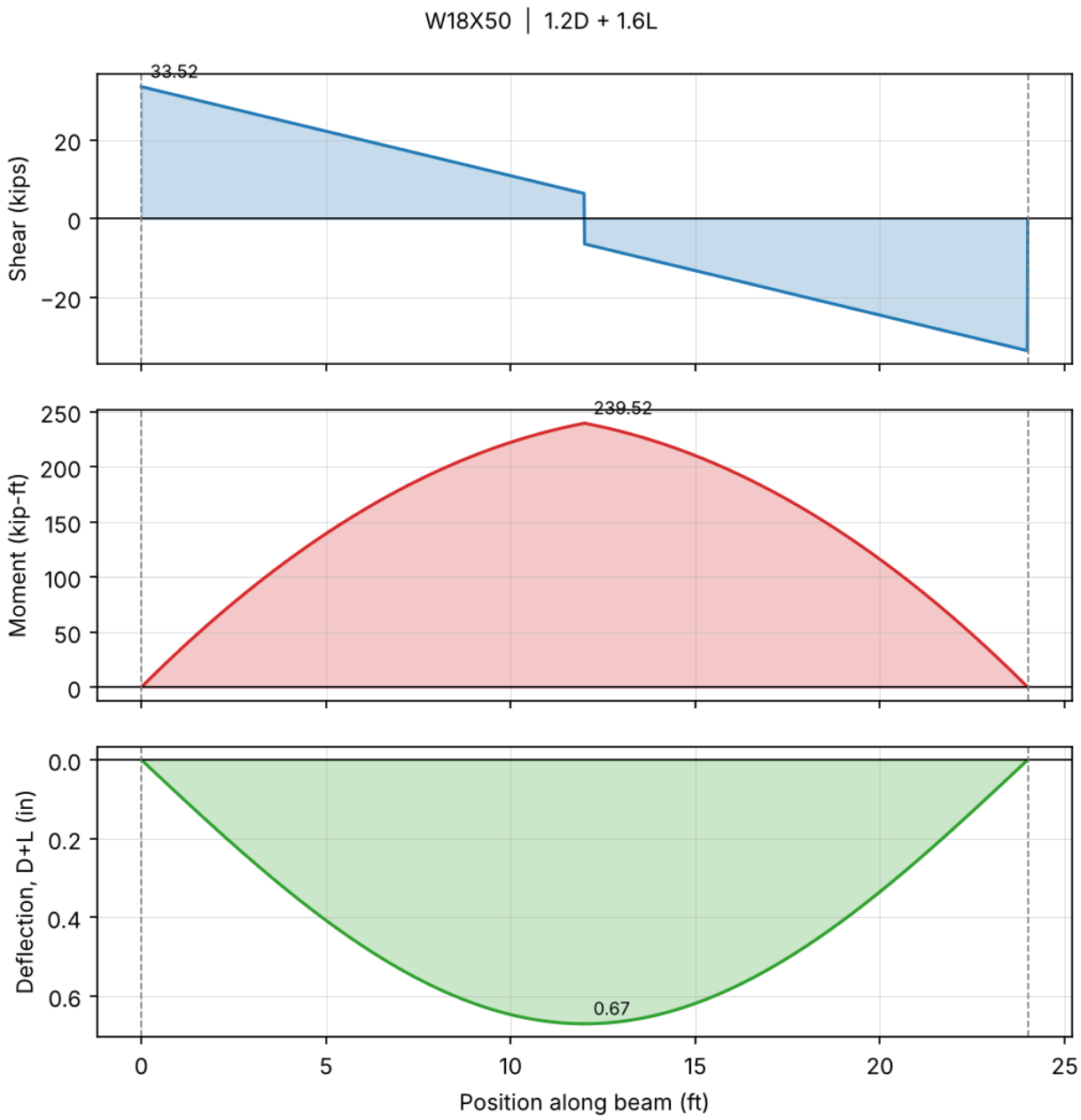
Load	Case	Magnitude	Location
Girder	L	8 kips	12 ft
Slab + deck	D	0.5 kip/ft	0 to 24 ft
Office live	L	1 kip/ft	0 to 24 ft
Self weight	D	0.050 kip/ft	0 to 24 ft

Section properties (AISC Shapes Database v16.0)

A	14.7 in²	d	18 in	bf	7.5 in	tf	0.57 in
tw	0.355 in	Ix	800 in^4	Sx	88.9 in³	Zx	101 in³
ry	1.65 in	rts	1.98 in	ho	17.4 in	J	1.24 in^4
bf/2tf	6.58	h/tw	45.2	Cw	3040 in^6	wt	50 lb/ft

Analysis

Reactions under 1.2D + 1.6L: R at 0 ft = 33.52 kips, R at 24 ft = 33.52 kips.
Mu,max = 239.52 kip-ft at 12.00 ft; Vu,max = 33.52 kips. Service deflection: live = 0.493 in, total = 0.670 in at 12.00 ft.


W18X50 | 1.2D + 1.6L
33.52
Shear (kips)
20
0
−20
239.52
Moment (kip-ft)
250
200
150
100
50
0
Deflection, D+L (in)
0.0
0.2
0.4
0.6
0.67
0
5
10
15
20
25
Position along beam (ft)

AISC 360-22 checks

Item	Value
Flange / web classification	compact / compact
Mp = Fy Zx	420.8 kip-ft
Lp (Eq. F2-5)	5.83 ft
Lr (Eq. F2-6)	16.95 ft
Mn, lateral-torsional buckling	420.8 kip-ft
Mn, flange local buckling	420.8 kip-ft
Mn (governing: yielding)	420.8 kip-ft
Aw = d tw	6.39 in²
Cv1	1.000
Vn = 0.6 Fy Aw Cv1	191.7 kips
Resistance / safety factors	phi_b = 0.90, Omega_b = 1.67; phi_v = 1.00, Omega_v = 1.50

Lb <= Lp: no LTB reduction (Eq. F2-1)
h/tw = 45.2 <= 2.24*sqrt(E/Fy) = 53.9: phi_v = 1.00, Cv1 = 1.0

Check	Demand	Capacity	Ratio	Status
Flexure (1.2D + 1.6L)	239.520 kip-ft	378.750 kip-ft	0.632	**OK**
Shear (1.2D + 1.6L)	33.520 kips	191.700 kips	0.175	**OK**
Live load deflection (L/360)	0.493 in	0.800 in	0.617	**OK**
Total load deflection (L/240)	0.670 in	1.200 in	0.559	**OK**

Narrative

The W18X50 (Fy = 50 ksi) is simply supported over 24 ft and was checked by LRFD using 1.2D + 1.6L. The maximum moment is 239.52 kip-ft at 12 ft and the maximum shear is 33.52 kips.

Flexural strength is governed by yielding with Mn = 420.8 kip-ft (Mp = 420.8 kip-ft, Lp = 5.83 ft, Lr = 16.95 ft, Lb = 0 ft). Nominal shear strength is 191.7 kips. Live load deflection is 0.493 in and total load deflection is 0.67 in.

The member passes. The governing check is Flexure (1.2D + 1.6L) at a demand to capacity ratio of 0.632.

Beam Check Desk is a student-built tool for checking hand calculations. It is not a substitute for engineering judgment or a licensed engineer's review, and it is not for use on real design work.